Macroeconomics I — MPE Insper

Practice list: Measurement, Welfare and Growth — qualifier level (Listas 1–2)

Solutions and grading rubric

Instructions. Multiple choice: mark the single correct option (no penalty for wrong answers). *True/False*: every 2 wrong items cancel 1 right one (score = $\max\{0, \text{right} - \text{wrong}/2\}\times$ item value); blank items do not count. Discursive questions: show every step. Numbers are illustrative and chosen to give clean answers.

Block 1 — Price indices, new goods, welfare and growth accounting

(Kurlat chs. 1–2, 5 • 25 points)

Question 1. (10 points) Objective items.

a) (3 points) A household has CES preferences with elasticity of substitution $\eta = 3$ between two goods and, at base prices, spends half of its budget on each. The price of good 1 rises 50% and the price of good 2 does not change. By how much does a Laspeyres index overstate the true change in the cost of living?

A. 2.1%

B. 6.2%

C. 25.0%

D. 0%

Answer: B

Step — exact index. For homothetic CES the cost of a util is $P = \left(\sum \omega_i p_i^{1-\eta}\right)^{1/(1-\eta)}$, so the exact index is the power mean of order $1-\eta$ of the price relatives with base expenditure weights (Question 2(b)): $\left(\frac{1}{2}1.5^{-2} + \frac{1}{2}\right)^{-1/2} = 0.7222^{-1/2} = 1.1767$.

Step — bias. Laspeyres $= \frac{1}{2}\cdot 1.5 + \frac{1}{2} = 1.25$; $1.25/1.1767 - 1 = 6.2\%$. Option A is the Cobb–Douglas answer ($\eta = 1$, geometric mean $\sqrt{1.5}$): the bias grows with substitutability. Second-order check: $\frac{\eta}{2}\operatorname{Var}(\ln R) = 1.5 \cdot 0.25(\ln 1.5)^2 = 0.062$.

Ref: Kurlat (2020), §1.2, pp. 22–25.

b) (3 points) *Welfare × growth.* Over 30 years Brazil's mean consumption grows 1.5% a year, life expectancy rises from 66 to 76 years, the variance of log consumption falls from 1.2 to 1.0, and leisure does not change. With the Kurlat–Jones–Klenow utility, $\sigma = 1$, $\bar{u} = 5$ and initial mean consumption normalized to one, the average annual growth of consumption-equivalent welfare is:

A. 1.50%

B. 1.83%

C. 4.33%

D. 2.30%

Answer: C

Step — indifference across dates. λ scales initial consumption to match final welfare: $a_0[\bar{u} + \ln\lambda + \mathbb{E}\ln c_0] = a_1[\bar{u} + \mathbb{E}\ln c_1]$, with $\mathbb{E}\ln c = \ln\bar{c} - v/2$ (lognormal): $\mathbb{E}\ln c_0 = -0.6$, $\mathbb{E}\ln c_1 = 0.45 - 0.5 = -0.05$.

Step — solve. $\ln\lambda = \frac{76}{66}(4.95) - 5 + 0.6 = 1.30$, i.e. $1.30/30 = 4.33\%$ a year: consumption 0.45, inequality 0.10, longevity $(76/66 - 1)\cdot 4.95 = 0.75$. B omits longevity; D adds $\ln(76/66)$ instead of weighting the value of life.

Ref: Kurlat (2020), §2.2, eqs. (2.2.1)–(2.2.2).

c) (2 points, T or F) A Paasche price index can never exceed the corresponding Laspeyres index.

Answer: F

Step — counterexample. Equal base shares, price relatives $(1.2, 1)$, quantity relatives $(1.3, 1)$ (demand shifts toward the good whose price rises): Laspeyres 1.100, Paasche $\frac{0.5\cdot1.2\cdot1.3+0.5}{0.5\cdot1.3+0.5} = 1.113$. In general $P/L = 1 + \mathrm{Cov}_w(R_p, R_q)/(\bar{R}_p\bar{R}_q)$ (Bortkiewicz): Paasche exceeds Laspeyres when price and quantity relatives covary positively.

Ref: Kurlat (2020), §1.2.

d) (2 points, T or F) *PPP × productivity growth.* Tradables obey the law of one price and the nominal exchange rate is constant. If Brazil's tradable-sector productivity grows 2 points a year faster than the US's while non-tradable productivity grows equally in both, Brazil's price level relative to the US rises by 2γ points a year, γ being the non-tradable share in consumption.

Answer: T

Step — Balassa–Samuelson in growth rates. $P = p_T(A_T/A_N)^\gamma$ in each country (common wage, zero profits), and $p_T^{BR} = Ep_T^{US}$; hence $d\ln\frac{P^{BR}}{EP^{US}} = \gamma[(g_T - g_N)^{BR} - (g_T - g_N)^{US}] = 2\gamma$ points.

Ref: Kurlat (2020), §1.2; Romer (2012), §4.2.

Question 2. (15 points) Discursive. *Measurement × growth accounting.* Consumers have CES preferences over the set Ω_t of varieties available at date t,

$$U = \Big(\sum_{i\in\Omega_t}(b_i x_i)^{\frac{\eta-1}{\eta}}\Big)^{\frac{\eta}{\eta-1}}, \qquad \eta > 1,$$

where $b_i > 0$ is a quality weight. The statistical agency observes prices and expenditure shares and builds its deflators from varieties present in both periods.

a) (3 points) Find the minimum expenditure needed to obtain one unit of utility at prices p, and each variety's expenditure share.

Step — problem. $\min\sum_i p_i x_i$ s.t. $\sum_i(b_i x_i)^\rho = 1$, $\rho \equiv \frac{\eta-1}{\eta} \in (0,1)$. FOC with multiplier μ: $p_i = \mu\rho b_i^\rho x_i^{\rho-1}$.

Step — demands. $x_i = (\mu\rho)^\eta b_i^{\eta-1}p_i^{-\eta}$, using $\frac{1}{1-\rho} = \eta$ and $\frac{\rho}{1-\rho} = \eta - 1$. The constraint gives $(\mu\rho)^{\eta-1}\sum_i(p_i/b_i)^{1-\eta} = 1$.

Step — cost of a util. Spending $= (\mu\rho)^\eta\sum_i b_i^{\eta-1}p_i^{1-\eta} = (\mu\rho)^\eta(\mu\rho)^{1-\eta} = \mu\rho$, so

$$P = \Big(\sum_i (p_i/b_i)^{1-\eta}\Big)^{\frac{1}{1-\eta}}, \qquad s_i = \frac{p_i x_i}{P} = \frac{(p_i/b_i)^{1-\eta}}{P^{1-\eta}}.$$

Homotheticity makes P independent of the utility level: it is a true price index.

Ref: Kurlat (2020), §1.2 (biases of price indices); formula sheet `rules/01`.

Grading rubric.

- FOC and demands: 1.5 pts.
- P and the shares: 1.5 pts. A formula without derivation: max. 1 pt in the item.

b) (3 points) Keep the set of varieties fixed between dates 0 and 1. Express the exact index P_1/P_0 in terms of observable price relatives $R_i = p_{i1}/p_{i0}$ and shares, rank it against the Laspeyres and Paasche indices, and approximate the Laspeyres bias to second order in the dispersion of $\ln R_i$.

Step — base weights. $(P_1/P_0)^{1-\eta} = \sum_i(p_{i1}/b_i)^{1-\eta}/P_0^{1-\eta} = \sum_i s_i^0 R_i^{1-\eta}$: the power mean of order $1-\eta$ with base weights, $M_{1-\eta}(R; s^0)$.

Step — current weights. $s_i^1 = s_i^0 R_i^{1-\eta}/\sum_j s_j^0 R_j^{1-\eta}$, so $\sum_i s_i^1 R_i^{\eta-1} = 1/\sum_j s_j^0 R_j^{1-\eta}$ and $P_1/P_0 = M_{\eta-1}(R; s^1)$.

Step — ranking. Laspeyres $= M_1(R; s^0)$, Paasche $= \big(\sum_i s_i^1/R_i\big)^{-1} = M_{-1}(R; s^1)$. Power means increase with the order; since $1-\eta < 1$ and $\eta - 1 > -1$: Paasche $\le P_1/P_0 \le$ Laspeyres, with equality iff all R_i are equal.

Step — second order. With $z_i = \ln R_i$, mean $\bar{z}$ and variance V (base weights), $\ln M_r = \frac{1}{r}\ln\sum s_i e^{rz_i} \approx \bar{z} + \frac{r}{2}V$. Hence $\ln L - \ln(P_1/P_0) \approx \frac{1}{2}V - \frac{1-\eta}{2}V = \frac{\eta}{2}V$.

Ref: Kurlat (2020), §1.2, pp. 22–25; Jones (2020), §2.3.

Grading rubric.

- Exact index with base and current weights: 1.5 pts.
- Ranking by the power-mean inequality: 1 pt. Ranking asserted: 0 for this line.
- Second-order bias: 0.5 pt.

c) (3 points) Now some varieties are new at date 1 and none disappears. Find the exact index in terms of the index over continuing varieties and observable shares. What happens as $\eta \to \infty$ and as $\eta \to 1$?

Step — split the index. Let C be the continuing set and $\lambda_t \equiv \frac{\sum_{i\in C}(p_{it}/b_i)^{1-\eta}}{\sum_{i\in\Omega_t}(p_{it}/b_i)^{1-\eta}}$, which by (a) is the expenditure share of continuing varieties at t ($\lambda_0 = 1$). Then $P_t^{1-\eta} = (P_t^C)^{1-\eta}/\lambda_t$.

Step — formula. Raising to $1/(1-\eta)$: $P_t = P_t^C \lambda_t^{1/(\eta-1)}$, so

$$\frac{P_1}{P_0} = \frac{P_1^C}{P_0^C}\Big(\frac{\lambda_1}{\lambda_0}\Big)^{\frac{1}{\eta-1}}.$$

Step — limits. $\eta \to \infty$: the exponent $\to 0$ and the correction disappears (perfect substitutes add nothing except through price). $\eta \to 1^+$: the exponent $\to \infty$ and, with $\lambda_1 < 1$, the correction $\to 0$: new varieties become infinitely valuable.

Ref: Kurlat (2020), §1.2 (new-goods bias).

Grading rubric.

- λ_t as an observable share: 1 pt.
- Formula derived: 1.5 pts.
- Both limits: 0.5 pt.

d) (3 points) Continuing-variety consumption inflation is 5% and new varieties take 4% of date-1 consumption spending; $\eta = 4$. New varieties appear only among consumption goods (share $s_C = 0.8$ of GDP); investment goods (share $s_I = 0.2$) have their own deflator bias b_I from unmeasured quality change. In a steady state with $\alpha = 1/3$, how is the Solow residual of growth accounting distorted? Compute it for $b_I = 0$, $b_I = b_C$ and $b_I = 3b_C$, where b_C is the consumption-deflator bias.

Step — consumption bias. True consumption inflation $1.05 \cdot 0.96^{1/3} - 1 = 3.58\%$: $b_C = 1.42$ points a year.

Step — measured growth. Real GDP growth is nominal growth minus the deflator, so it is understated by $b_Y = s_C b_C + s_I b_I$. Capital is built by perpetual inventory from deflated investment, so in a steady state its measured growth is understated by b_I.

Step — residual. $g_A^m = (g_Y - b_Y) - \alpha(g_K - b_I) - (1-\alpha)g_L$, hence

$$g_A^m - g_A = -(s_C b_C + s_I b_I) + \alpha b_I.$$

$b_I = 0$: -1.14 points; $b_I = b_C$: $-(1-\alpha)b_C = -0.95$; $b_I = 3b_C$: -0.57.

Economic intuition. Understated output lowers measured TFP growth; understated capital raises it, because the residual absorbs whatever inputs miss. Quality change in equipment is the embodied technical progress of Romer's problem 1.12, which growth accounting books as TFP.

Ref: Kurlat (2020), §5.4, eq. (5.4.2), p. 87; Romer (2012), §1.7 and problem 1.12.

Grading rubric.

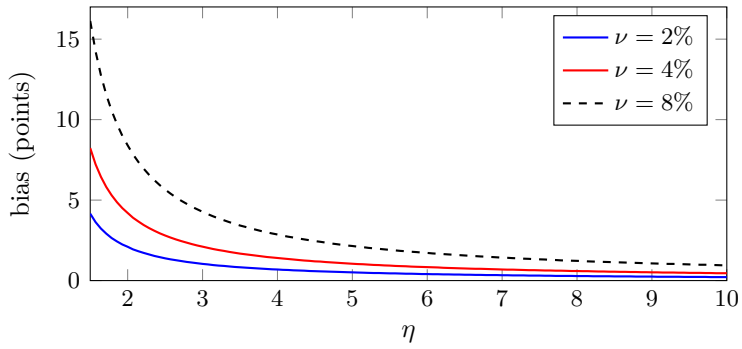
- b_C: 0.5 pt.
- Biases of measured output and capital growth: 1 pt.
- Residual formula and the three numbers: 1.5 pts. Ignoring the capital channel: max. 1.5 pts in the item.

e) (3 points) *Parameter analysis.* How do the inflation bias and the residual bias depend on η, the new-variety share $\nu \equiv 1 - \lambda_1$ and b_I? Prove the signs, fill the inflation bias for $\eta \in \{2, 4, 8\}$ and $\nu \in \{2\%, 4\%, 8\%\}$, find the b_I at which the residual bias changes sign, and draw the inflation bias against η for the three values of ν.

Step — signs. Bias $B(\eta, \nu) = 1.05\,[1 - (1-\nu)^{1/(\eta-1)}]$. Since $\ln(1-\nu) < 0$, $(1-\nu)^{1/(\eta-1)}$ rises with η and falls with ν: $\partial B/\partial \eta < 0$, $\partial B/\partial \nu > 0$. The residual bias $-(s_C b_C) + (\alpha - s_I) b_I$ rises with b_I iff $\alpha > s_I$, and is zero at $b_I^* = s_C b_C/(\alpha - s_I) = 8.5$ points: only implausibly fast equipment quality change would make measured TFP growth too high.

Step — table. Inflation bias in points (continuing inflation 5%):

	$\nu = 2\%$	$\nu = 4\%$	$\nu = 8\%$
$\eta = 2$	2.10	4.20	8.40
$\eta = 4$	0.70	1.42	2.88
$\eta = 8$	0.30	0.61	1.24



Step — limits and reading. $\eta \to 1$: the bias tends to the whole of continuing inflation plus more (the index collapses toward zero); $\eta \to \infty$: no bias. The bias is roughly proportional to ν and to $1/(\eta - 1)$: it is a problem for goods with many new, differentiated varieties, not for commodities.

Ref: Kurlat (2020), §1.2 and §5.4.

Grading rubric.

- Signs proved: 1 pt. Signs listed without the argument: max. 0.5 pt here.
- Table: 0.5 pt.
- b_I^* and its reading: 0.5 pt.
- Graph (axes, three decreasing convex curves ordered by ν): 1 pt.

Block 2 — Growth, capital flows and national income

(Kurlat chs. 4–5; Romer chs. 1, 4 • 25 points)

Question 3. (10 points) Objective items.

a) (3 points) *Development accounting × capital flows.* Let $Y = AK^{\alpha}(hL)^{1-\alpha}$ with $\alpha = 0.4$ and the same A in Brazil and the US. Brazil's output per worker is 25% of the US's and its human capital per worker 60%. The ratio of marginal products of capital (Brazil/US) implied by the closed-economy model is about:

A. 8.0

B. 3.7

C. 1.0

D. 2.5

Answer: B

Step — MPK in terms of observables. $MPK = \alpha y/k$ and $k = (y/(Ah^{1-\alpha}))^{1/\alpha}$, so $MPK = \alpha A^{1/\alpha}(y/h)^{(\alpha-1)/\alpha}$.

Step — ratio. $(0.25/0.6)^{-1.5} = 3.7$. Without human capital, $0.25^{-1.5} = 8.0$ (A). Capital should flow to Brazil; that it does not suggests lower A (Lucas paradox).

Ref: Kurlat (2020), §5.3 (the capital accumulation hypothesis), pp. 80–86.

b) (3 points) With $f(k) = [\alpha k^{-1} + 1 - \alpha]^{-1}$ (elasticity of substitution 1/2), $\alpha = 1/2$ and $n + g + \delta = 0.08$, the Golden Rule saving rate is:

A. 50%

B. 20%

C. 8%

D. 32%

Answer: B

Step — Golden Rule. $f(k) = \frac{k}{\alpha+(1-\alpha)k}$, $f'(k) = \frac{\alpha}{(\alpha+(1-\alpha)k)^2} = 0.08 \Rightarrow \alpha + (1-\alpha)k = 2.5$, $k = 4$, $f = 1.6$.

Step — saving rate. $s = \frac{(n+g+\delta)k}{f(k)} = \frac{0.32}{1.6} = 20\%$, which equals the capital share *at* k_{gr}, not α (A).

Ref: Kurlat (2020), §4.3; Romer (2012), §1.8 (CES).

c) (2 points, T or F) With CES production and an elasticity of substitution different from one, a balanced growth path with a constant capital–output ratio exists only if technical progress is labor-augmenting.

Answer: T

Step — Uzawa. On a balanced path K and Y grow at the same rate. With $Y = F(BK, AL)$ CES, $Y/K = [\alpha B^\rho + (1-\alpha)(AL/K)^\rho]^{1/\rho}$ is constant along an exact balanced path only if B is constant and AL/K is constant, i.e. technology augments labor. Cobb–Douglas is the exception, where all forms are equivalent.

Ref: Romer (2012), §1.2 and problem 1.12.

d) (2 points, T or F) *Growth × open economy.* In a small open Solow economy with perfect capital mobility, a permanent rise in the saving rate permanently raises GDP per worker.

Answer: F

Step — why. GDP per effective worker is pinned by $f'(k) = r^* + \delta$; saving changes who owns the capital (net foreign assets and national income), not domestic production (Question 4).

Ref: Romer (2012), problem 4.4.

Question 4. (15 points) Discursive. *Growth × national accounts × capital flows.* Brazil as a small open economy. Output is $Y = K^\alpha(AhL)^{1-\alpha}$; capital moves freely and earns the world rate r^* net of depreciation δ; A grows at g, L at n, h is constant. Residents own wealth W (capital at home or abroad) and save a fraction s of gross national income (wages plus the gross return on wealth): $\dot{W} = s\,\text{GNI} - \delta W$. Numbers: $\alpha = 1/3$, $r^* = 0.04$, $\delta = 0.05$, $n + g = 0.03$.

a) (3 points) Characterize GDP per effective worker and its dynamics. What does a rise in s do to GDP? What does this imply for the Lucas paradox?

Step — capital is pinned. Firms rent capital at $r^* + \delta$: $f'(k) = \alpha k^{\alpha-1} = r^* + \delta$ with $k = K/(AhL)$, so $k^* = \left(\frac{\alpha}{r^*+\delta}\right)^{1/(1-\alpha)}$ at every date: capital flows in or out instantly.

Step — GDP. $y^* = k^{*\alpha}$ is constant, so GDP per worker Ahy^* grows at g from the start: no transition, and s does not enter. A higher s changes ownership of capital, not production.

Step — Lucas paradox. With equal A and h, the closed-economy MPK gap would be several-fold (item 3(a)) and capital would pour in. Small observed flows mean that A and h, which set k^* through Ah, differ.

Ref: Romer (2012), problem 4.4; Kurlat (2020), §5.3.

Grading rubric.

- k^* from the world rate: 1 pt.
- No transition and no role for s in GDP: 1 pt.
- Lucas-paradox reading: 1 pt.

b) (3 points) Derive the law of motion of wealth per effective worker, its steady state and the condition for stability.

Step — national income. Wages per effective worker are $(1-\alpha)y^*$; wealth earns $r^*+\delta$ gross. With $\omega \equiv W/(AhL)$: $\text{GNI}/(AhL) = (1-\alpha)y^* + (r^*+\delta)\omega$.

Step — law of motion. $\dot{\omega}/\omega = \dot{W}/W - n - g$ gives

$$\dot{\omega} = s(1-\alpha)y^* - \Lambda\,\omega, \qquad \Lambda \equiv n + g + \delta - s(r^*+\delta).$$

Step — solution. $\omega(t) = \omega^* + (\omega(0) - \omega^*)e^{-\Lambda t}$, $\omega^* = \frac{s(1-\alpha)y^*}{\Lambda}$: it converges iff $\Lambda > 0$. If $\Lambda \le 0$ wealth per effective worker grows without bound and the economy eventually ceases to be small.

Ref: Romer (2012), §1.3 (same method).

Grading rubric.

- GNI per effective worker: 1 pt.
- Law of motion: 1 pt. Omitting the return on wealth: max. 1 pt in the item.
- Explicit solution and stability: 1 pt.

c) (3 points) Find the saving rate above which Brazil is a net creditor in the steady state and interpret it. Find GNI/GDP in the steady state in closed form.

Step — creditor condition. Net foreign assets are $\omega^* - k^*$. Using $(r^*+\delta)k^* = \alpha y^*$: $\omega^* > k^* \iff s(1-\alpha)y^* > k^*\Lambda = k^*(n+g+\delta) - s\alpha y^* \iff s\,y^* > (n+g+\delta)k^*$, i.e.

$$s > \bar{s} \equiv \frac{(n+g+\delta)k^*}{y^*} = \frac{\alpha(n+g+\delta)}{r^*+\delta},$$

the closed-economy saving rate whose steady state is k^*.

Step — GNI/GDP. $\frac{\text{GNI}}{\text{GDP}} = (1-\alpha) + \alpha\frac{\omega^*}{k^*}$ and $\frac{\omega^*}{k^*} = \frac{s(1-\alpha)(r^*+\delta)}{\alpha\Lambda}$, so

$$\frac{\text{GNI}}{\text{GDP}} = (1-\alpha)\Big[1 + \frac{s(r^*+\delta)}{\Lambda}\Big] = \frac{(1-\alpha)(n+g+\delta)}{\Lambda},$$

which equals 1 exactly at $s = \bar{s}$: creditor status and GNI above GDP are the same statement.

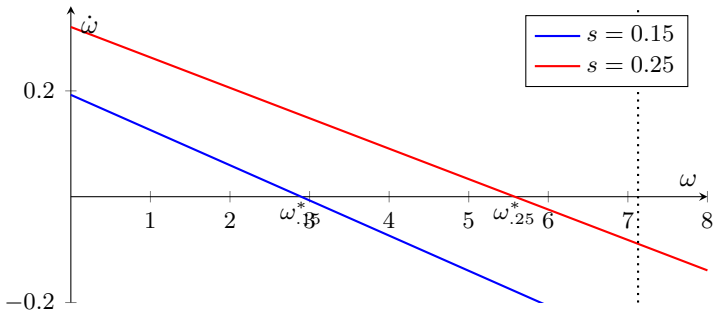
Ref: Romer (2012), §1.5 and problem 4.4.

Grading rubric.

- Creditor condition derived: 1.5 pts.
- GNI/GDP in closed form and the link at $\bar{s}$: 1.5 pts.

d) (3 points) With $s = 0.15$ compute k^*, y^*, ω^*, net foreign assets, the share of domestic capital owned abroad, GNI/GDP and the half-life of wealth. Draw $\dot{\omega}$ against ω for $s = 0.15$ and $s = 0.25$, marking k^*, and describe the transition after s rises.

Step — numbers. $k^* = (\frac{1/3}{0.09})^{3/2} = 7.128$, $y^* = 1.925$, wages 1.283; $\Lambda = 0.0665$; $\omega^* = 0.1925/0.0665 = 2.894$; net foreign assets -4.234 (59% of domestic capital foreign-owned); GNI/GDP $= 0.0533/0.0665 = 0.802$; half-life $\ln 2/0.0665 = 10.4$ years.



Step — transition. Each line crosses zero at its steady state and slopes down (slope $-\Lambda$): stable. After the rise in s, ω climbs from 2.894 toward 5.578 (half-life $\ln 2/0.0575 = 12.1$ years), still left of k^*: GNI rises gradually while GDP does not move.

Ref: Romer (2012), problem 4.4.

Grading rubric.

- Numbers: 1.5 pts.
- Graph: axes, two downward lines with the right intercepts and zeros, k^* marked: 1 pt.
- Transition read off the graph: 0.5 pt.

e) (3 points) *Parameter analysis.* How do ω^*/k^*, GNI/GDP and the half-life of wealth respond to s, r^*, $n+g$ and α? Prove the signs, fill the table for the baseline and for $s = 0.25$, $r^* = 0.06$, $n+g = 0.02$, $\alpha = 0.4$ (one change at a time), and give the limits as $s \to 0$ and as s approaches its upper bound.

Step — signs. From (c), $\frac{\omega^*}{k^*} = \frac{s(1-\alpha)(r^*+\delta)}{\alpha\Lambda}$, $\frac{\text{GNI}}{\text{GDP}} = \frac{(1-\alpha)N}{\Lambda}$, half-life $= \frac{\ln 2}{\Lambda}$, with $N = n+g+\delta$ and $\Lambda = N - s(r^*+\delta)$. Λ falls with s and r^* and rises with N; it does not depend on α. Hence: all three rise with s and with r^* (a higher return compounds wealth faster); ω^*/k^* and GNI/GDP fall with $n+g$ ($\partial[N/(N-a)]/\partial N = -a/(N-a)^2 < 0$) and so does the half-life; ω^*/k^* and GNI/GDP fall with α (labor income, the source of saving, is a smaller share), while the half-life does not depend on α.

Step — table.

	ω^*/k^*	GNI/GDP	half-life (years)
baseline	0.406	0.802	10.4
$s = 0.25$	0.783	0.928	12.1
$r^* = 0.06$	0.520	0.840	10.9
$n+g = 0.02$	0.478	0.826	12.3
$\alpha = 0.4$	0.305	0.722	10.4

Step — limits. $s \to 0$: $\omega^* \to 0$ and GNI/GDP $\to 1-\alpha$ (residents own nothing and earn only wages). $s \to N/(r^*+\delta)$ ($= 0.89$ here): $\Lambda \to 0$, wealth, GNI/GDP and the half-life explode; the small-economy assumption fails before.

Economic intuition. Openness separates where capital works (set by productivity and the world rate) from who owns it (set by saving); every parameter that raises the return on owned wealth relative to dilution by growth makes residents richer without changing what Brazil produces.

Ref: Romer (2012), problem 4.4; Kurlat (2020), §1.1 (GDP vs GNI).

Grading rubric.

- Signs proved from the closed forms: 1.5 pts.
- Table: 1 pt.
- Limits: 0.5 pt.